Evolution of Cosmic Structure

Lecture 15 – Cosmic Star formation/galaxy classes

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Learning Outcomes: I

What do we know of the first galaxies and the history of cosmic star formation? What effect has the development of galaxies had on the rest of the baryonic matter?

- Observations of high redshift galaxies
- Reionization – the impact of star formation on the intergalactic medium
- The history of star formation and the growth of stellar mass in the Universe

Cosmic star formation history

Various probes of the global SF rate: ρ_* (z) $M_\odot$ yr^{-1} comoving Mpc^{-3}

- Rest frame UV continuum flux (dominated by O & B stars)
- Hα and [O II] emission in spectroscopic surveys (O stars)
- mid-IR dust emission
- 1.4GHz radio emission

In general these SF indicators are sensitive mostly to the presence of *massive* stars. Hence estimate of the <u>total</u> SFR requires the assumption of some particular form for the *Initial Mass Functi*on of stars formed.

There is no simple `best method' : each has pros and cons (e.g. dust extinction, sample depth, z range and physical calibration uncertainties)

Each method has different time-sensitivity to main sequence activity, so if the SFR is not uniform, they will not generally give the same values for SFR.

Review paper: Madau & Dickinson (2014)

Cosmic star formation history

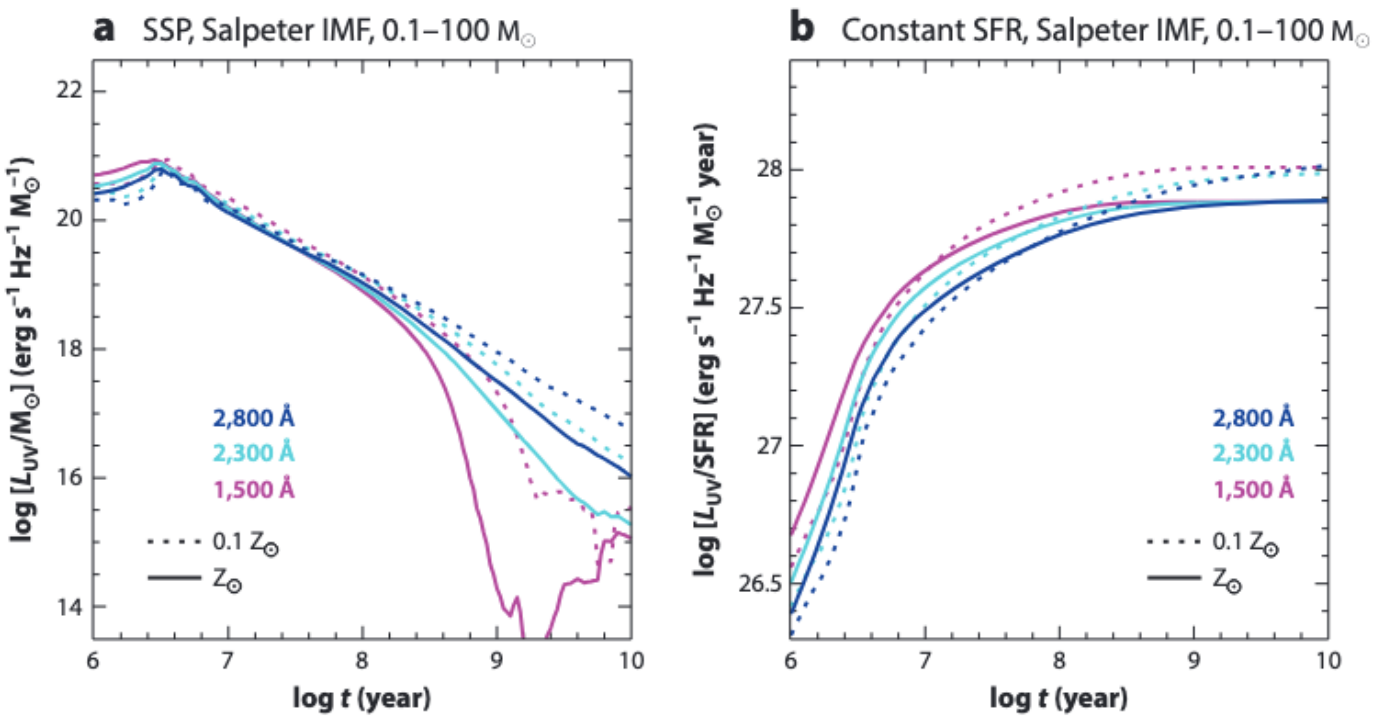
Kennicutt 1998 Ann Rev A&A 36, 189

1. **UV continuum (1250-2500 Å) :**

   Pro: Extensive datasets over 0<z<6: easily calibrated via MS models
   $M>5M_\odot$, timescales $>10^8$ yr

   Con: dust obscuration (several mag); IMF-dependent; Metallicity-dependent

$$\mathrm{SFR}(M_\odot\ \mathrm{year}^{-1}) = 1.4 \times 10^{-28} L_\nu(\mathrm{ergs\ s}^{-1}\ \mathrm{Hz}^{-1}).$$



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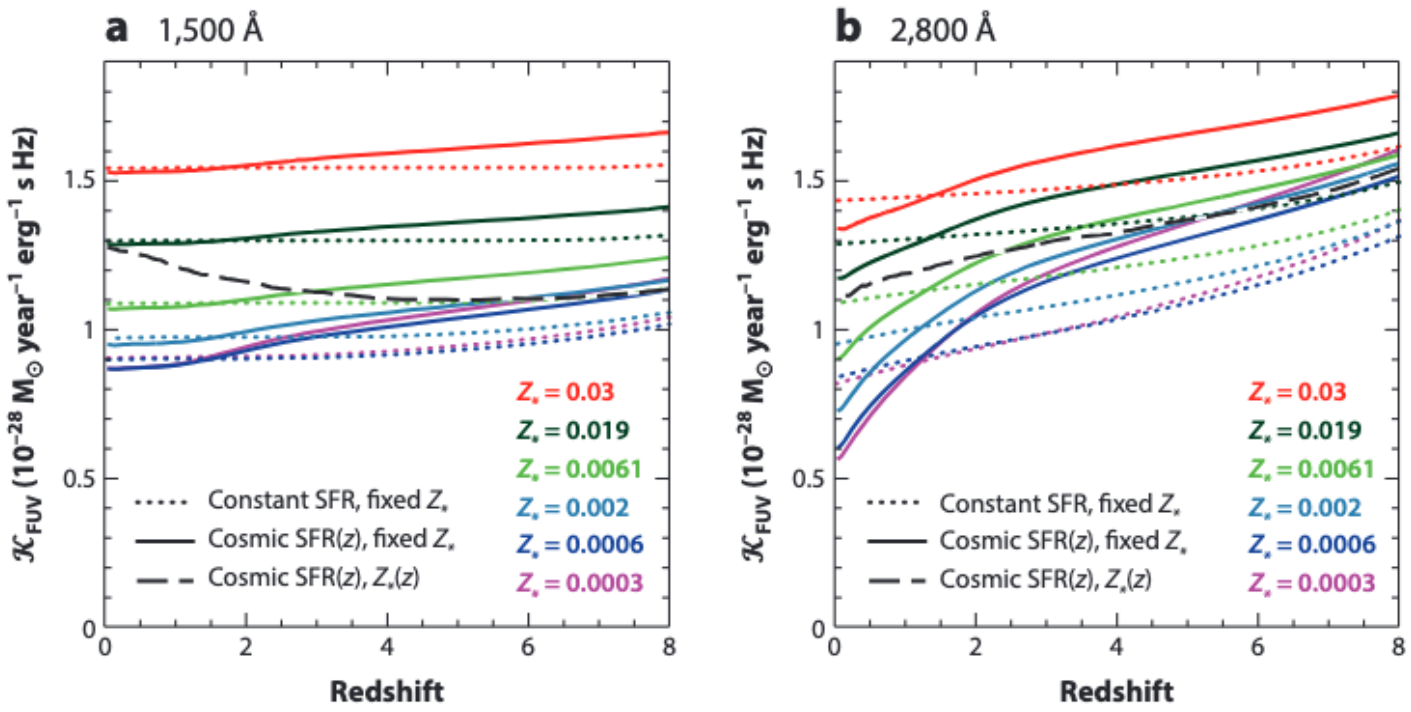

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2. **Line emission (Hα, [O II] :**
   Pro: Very sensitive probe, available to z~2: $M>10M_\odot$timescales $<10^6$ yr,
   Con: uncertain escape fraction of ionizing photons; strong IMF-dependence (×3), excitation uncertainties [OII]

$$\mathrm{SFR}(M_\odot\ \mathrm{year}^{-1}) = 7.9 \times 10^{-42}\ L(\mathrm{H}\alpha)\ (\mathrm{ergs\ s}^{-1})$$
$$\mathrm{SFR}(M_\odot\ \mathrm{year}^{-1}) = (1.4 \pm 0.4) \times 10^{-41}\ L[\mathrm{OII}]\ (\mathrm{ergs\ s}^{-1}),$$

3. **Far IR emission (10-300 μm) :**
   Pro: Independent method, available for obscured sources to high z:
   Con: uncertain source of dust heating (AGN/SF?); age of stellar pop, primarily applicable in starbursts, bolometric FIR flux required

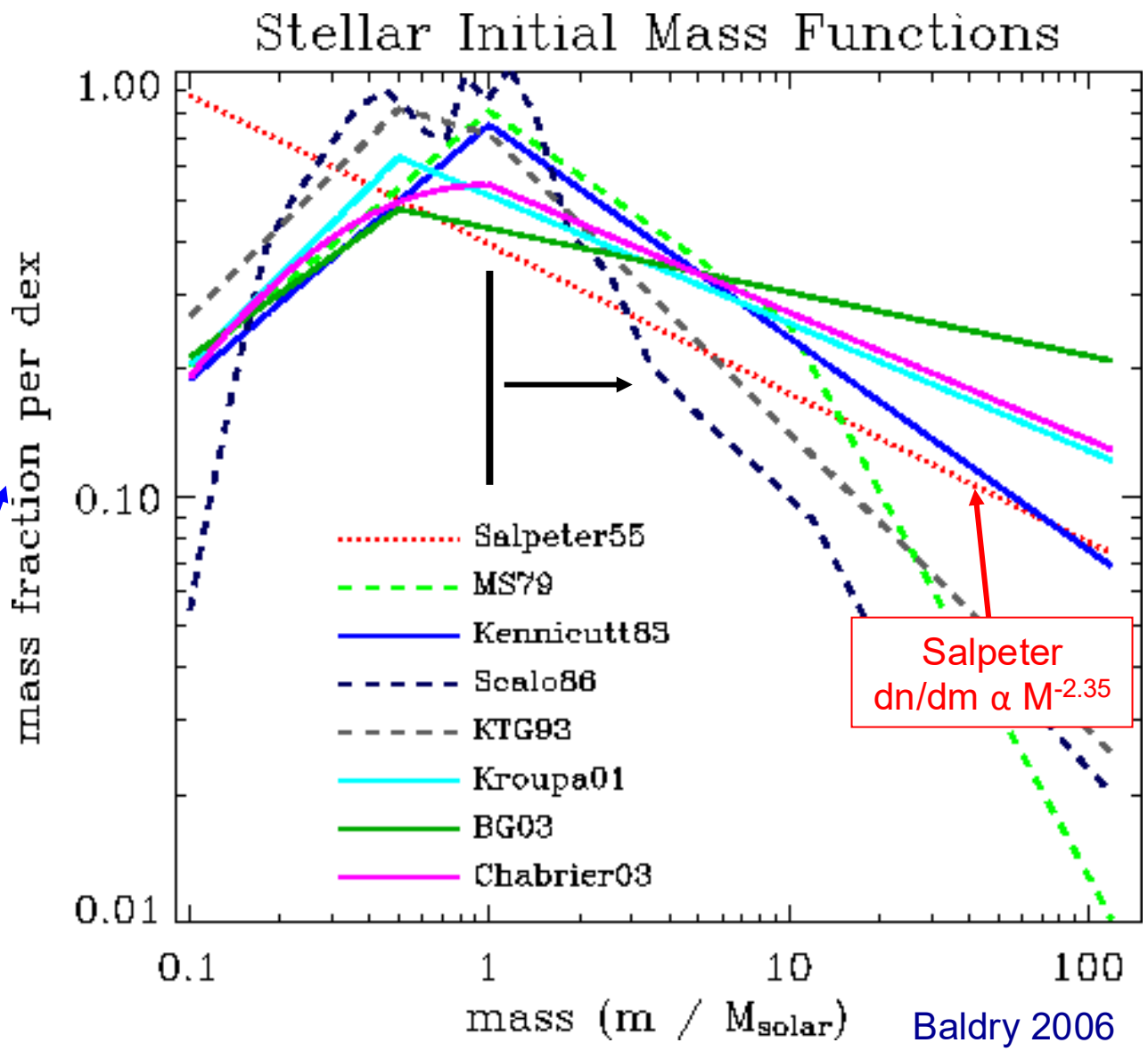
$$\mathrm{SFR}(M_\odot\ \mathrm{year}^{-1}) = 4.5 \times 10^{-44}\ L_{FIR}\ (\mathrm{ergs\ s}^{-1})\ (\mathrm{starbursts}),$$

Some Initial Mass Functions

Salpeter (1955) IMF at > 1 $M_\odot$ reproduces colors and Hα properties of spirals

IMF at < 1 $M_\odot$ makes minor contribution to light but is very important for mass inventory

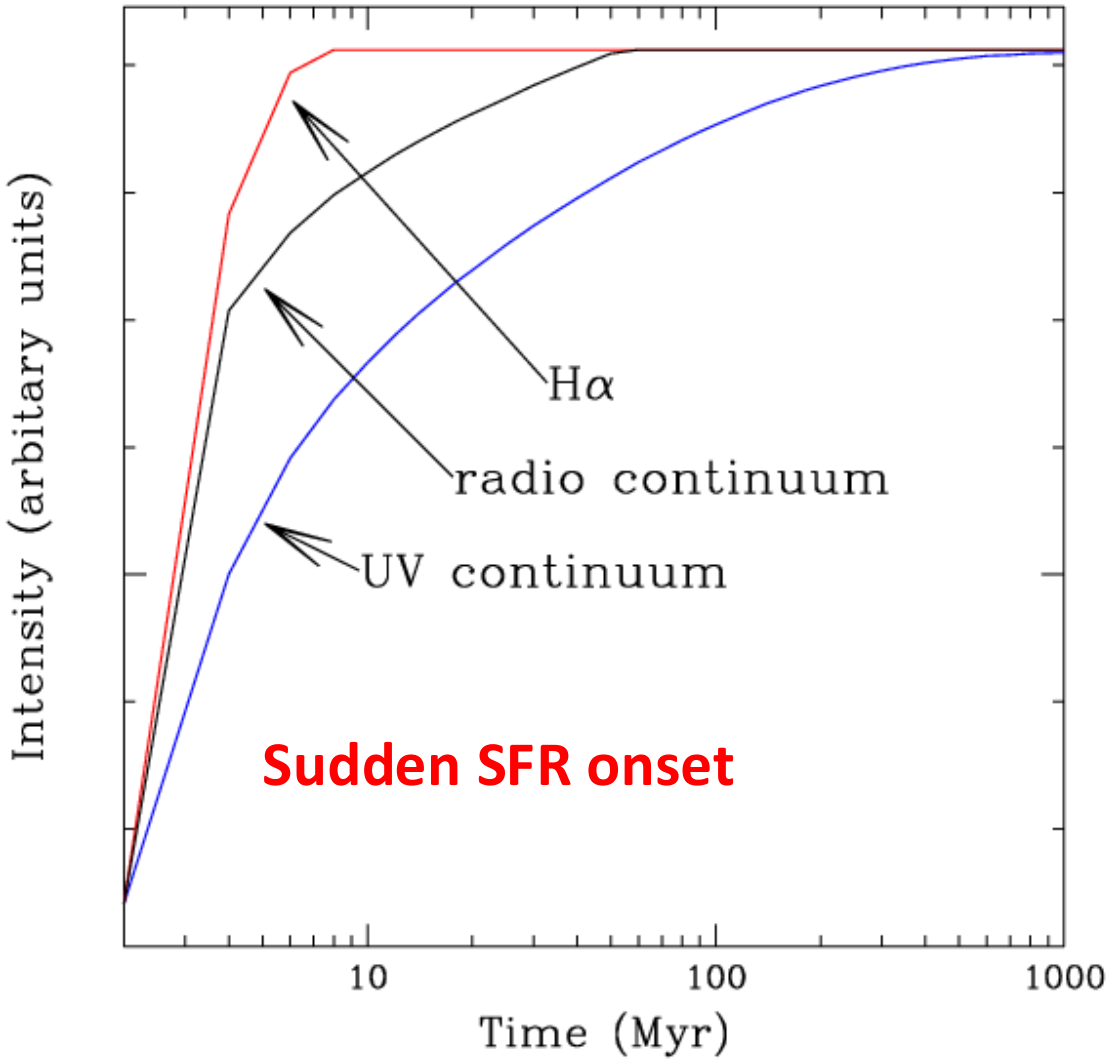
i.e. mass fraction per decade in mass



Baldry 2006

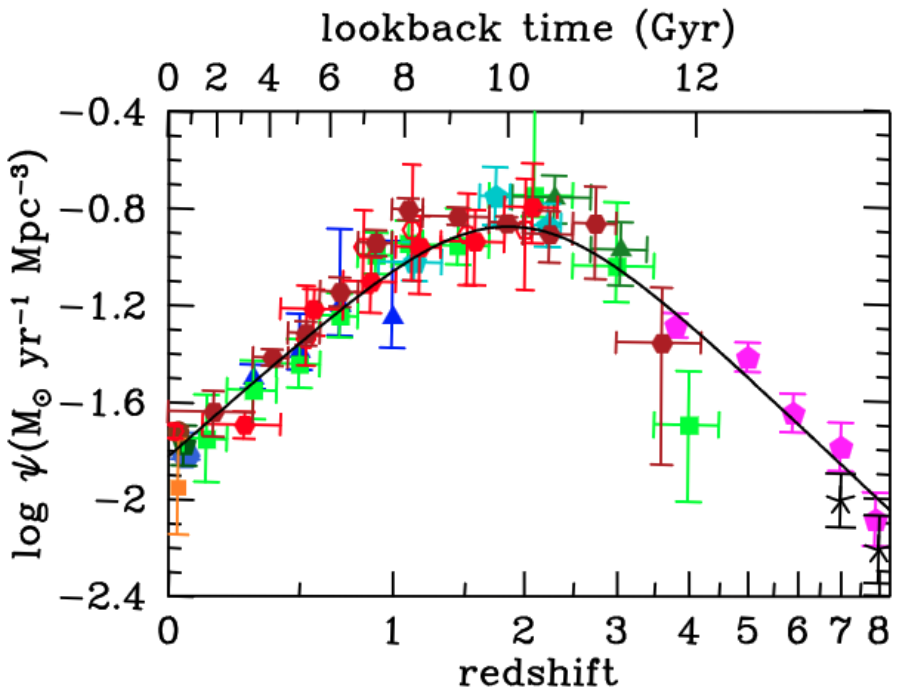
Time-dependence of SF diagnostics

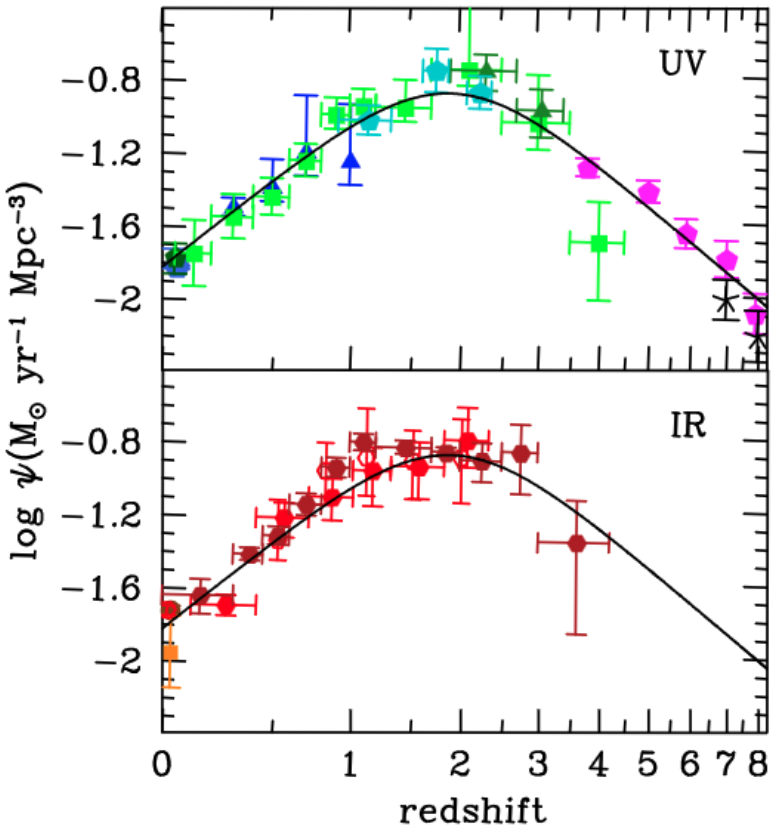
- Each SF diagnostic arises from a component of the stellar population whose lifetime is different, e.g. in a situation where the SF is erratic.
- Radio continuum is thought to arise from SN remnants and offers the potential of a dust-free diagnostic



Cosmic star formation history

Collecting results from many studies, including recent ones pushing out to z~8, the history of cosmic star formation shows a steep rise to *z*~2 and then a decline at higher *z*.



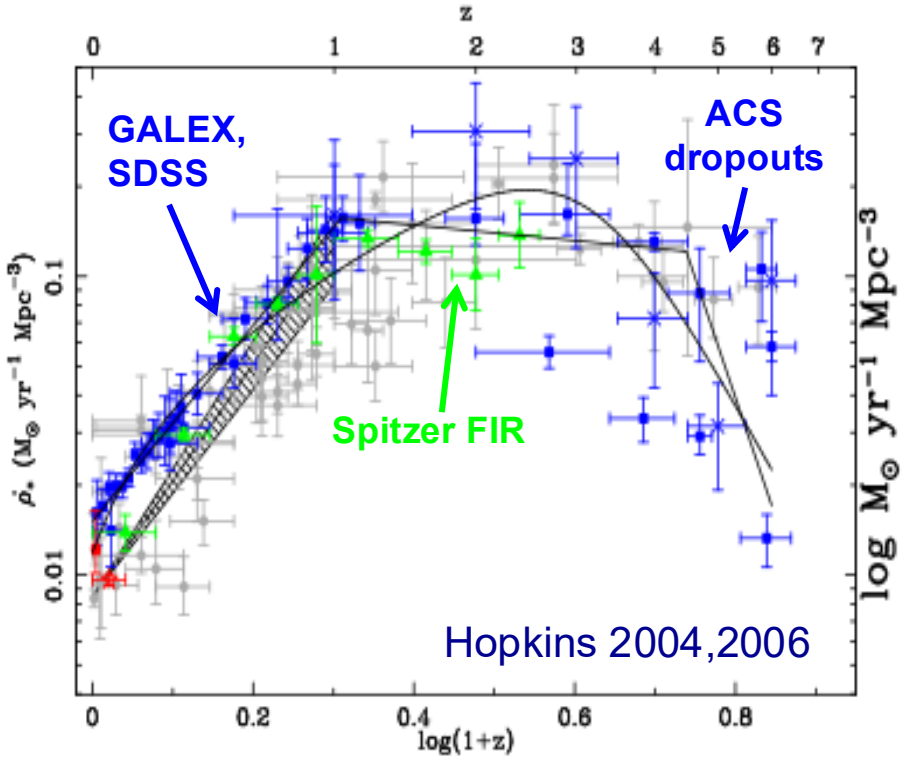


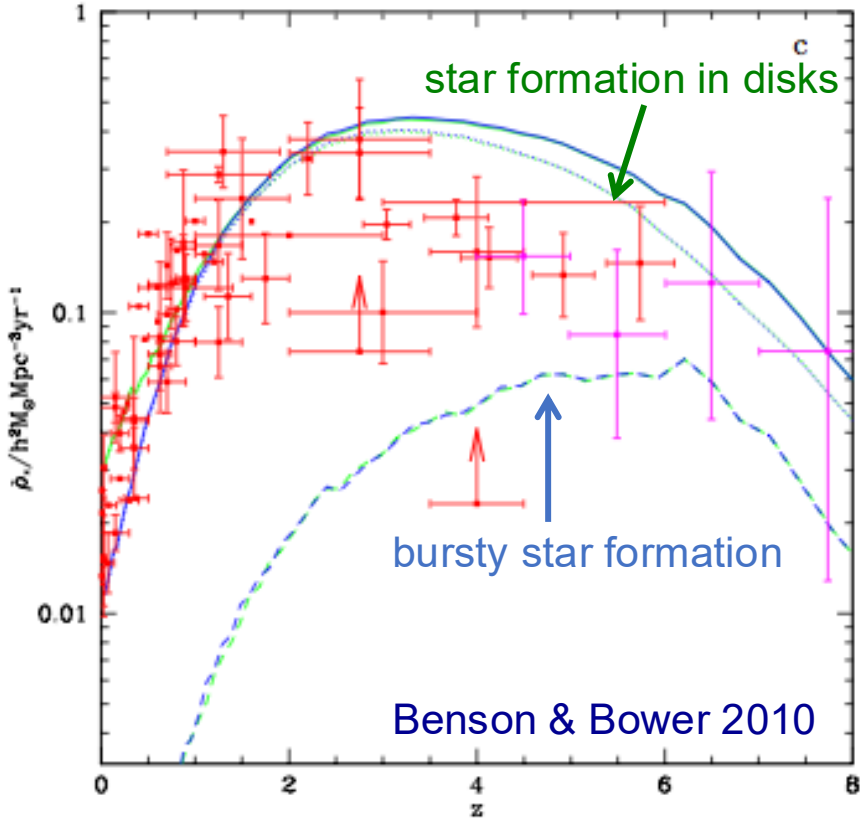
Madau & Dickinson 2014

Cosmic star formation history

Can models reproduce this evolution in the cosmic SFR?

An example, is the semi-analytic model of Benson & Bower, in which most of the SF is found to take place fairly steadily within galaxy disks. These authors note that the model is in "excellent agreement" with the data at z>1….





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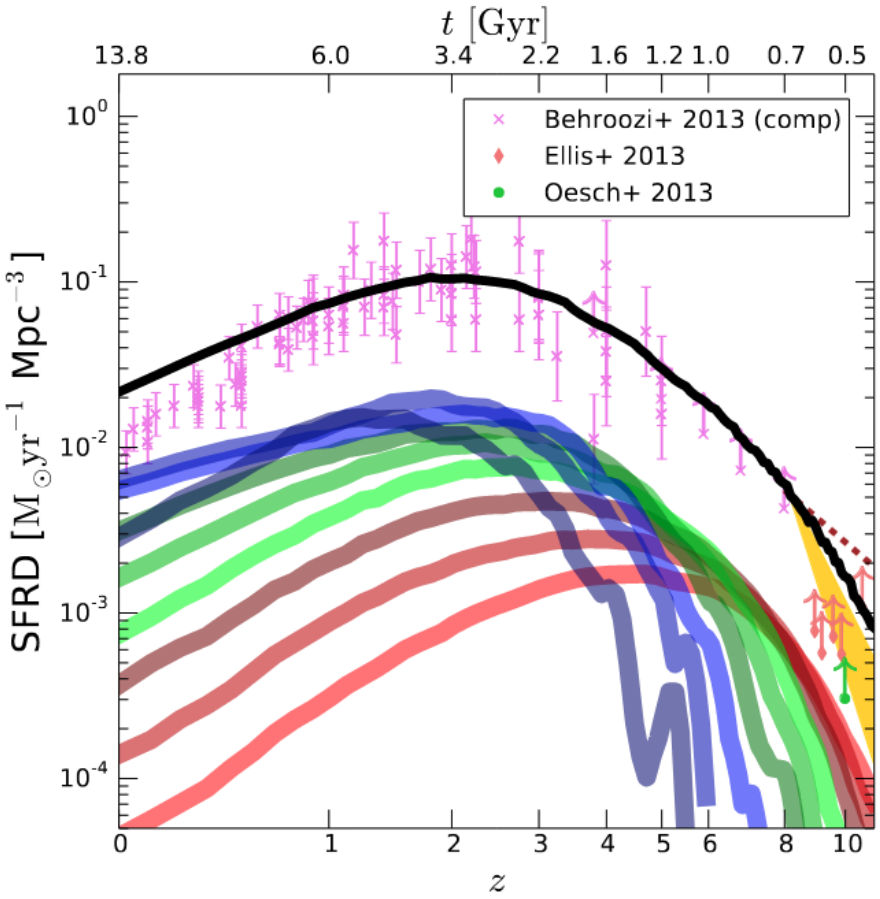
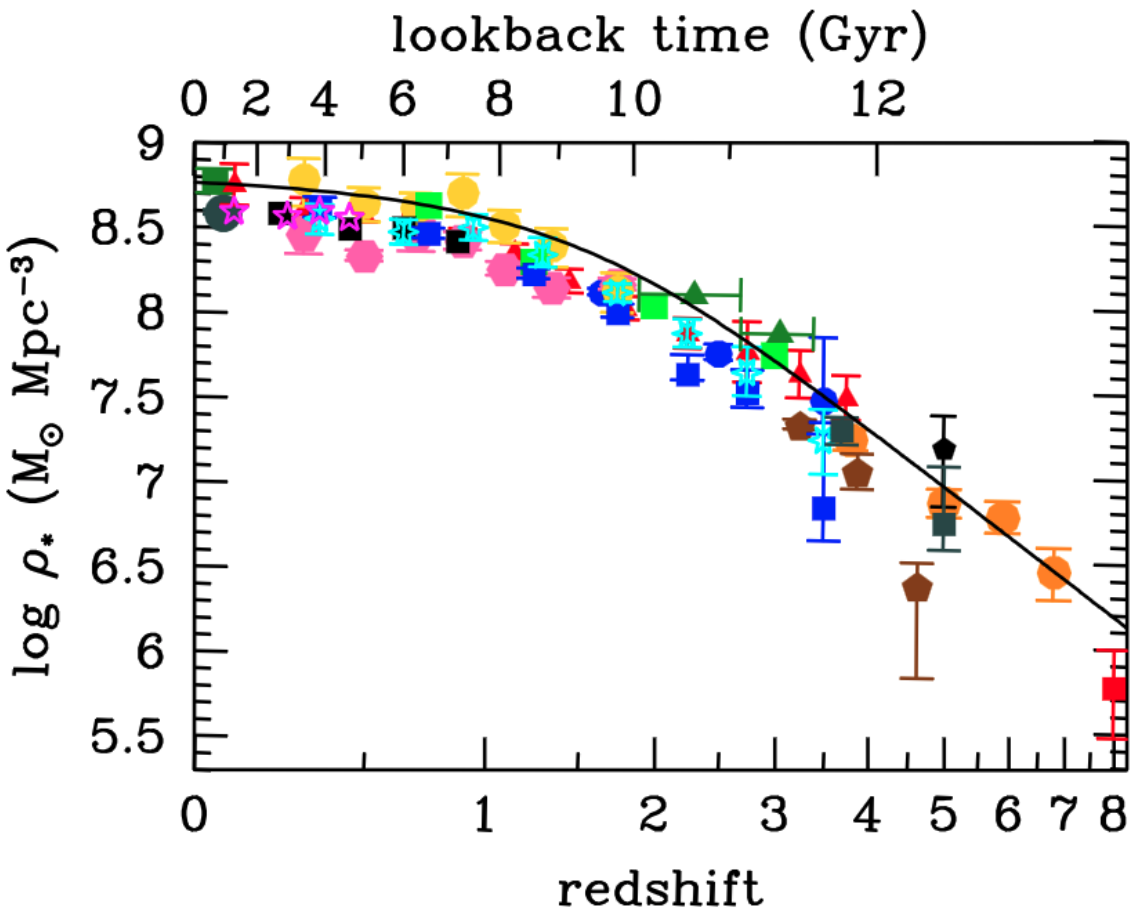


Stellar mass assembly

Integrating the stellar mass function over galaxies in the range 10^8-10^{13} $M_\odot$ allows the comoving density of stars to be estimated as a function of redshift, showing the build-up in stellar mass which results from cosmic star formation.

The IMF and extinction effects are the main uncertainties in these calculations.

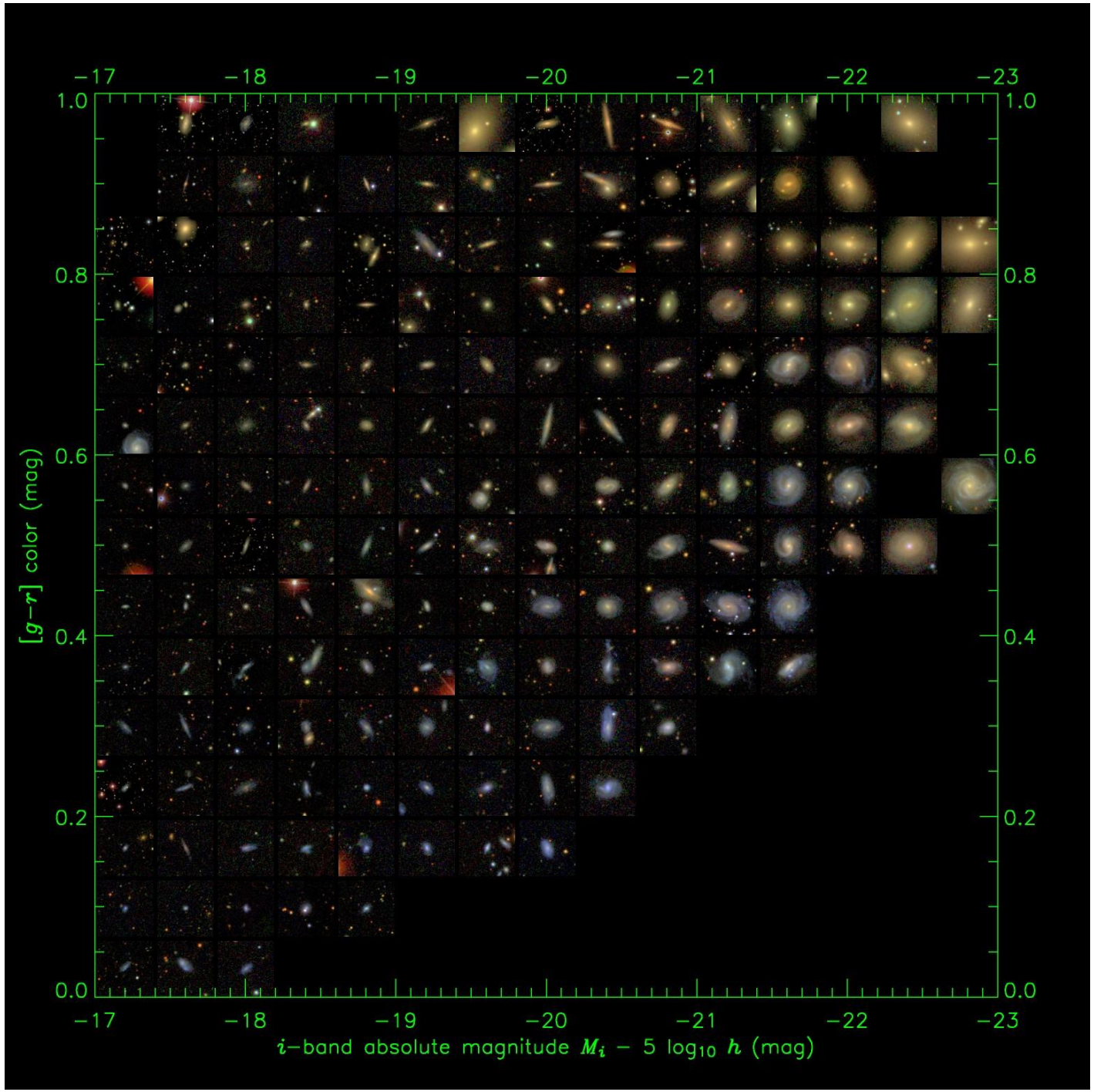


Madau & Dickinson 2014

Galaxy Properties and Classes

How can the zoo of galaxies be usefully classified, and what key physical characteristics differentiate the main classes? What is the distribution of galaxy luminosities, and how does this relate to the dark halo masses? What is the impact of hierarchical structure development on galaxies? What are the main scaling properties of galaxies, and what do they tell us?

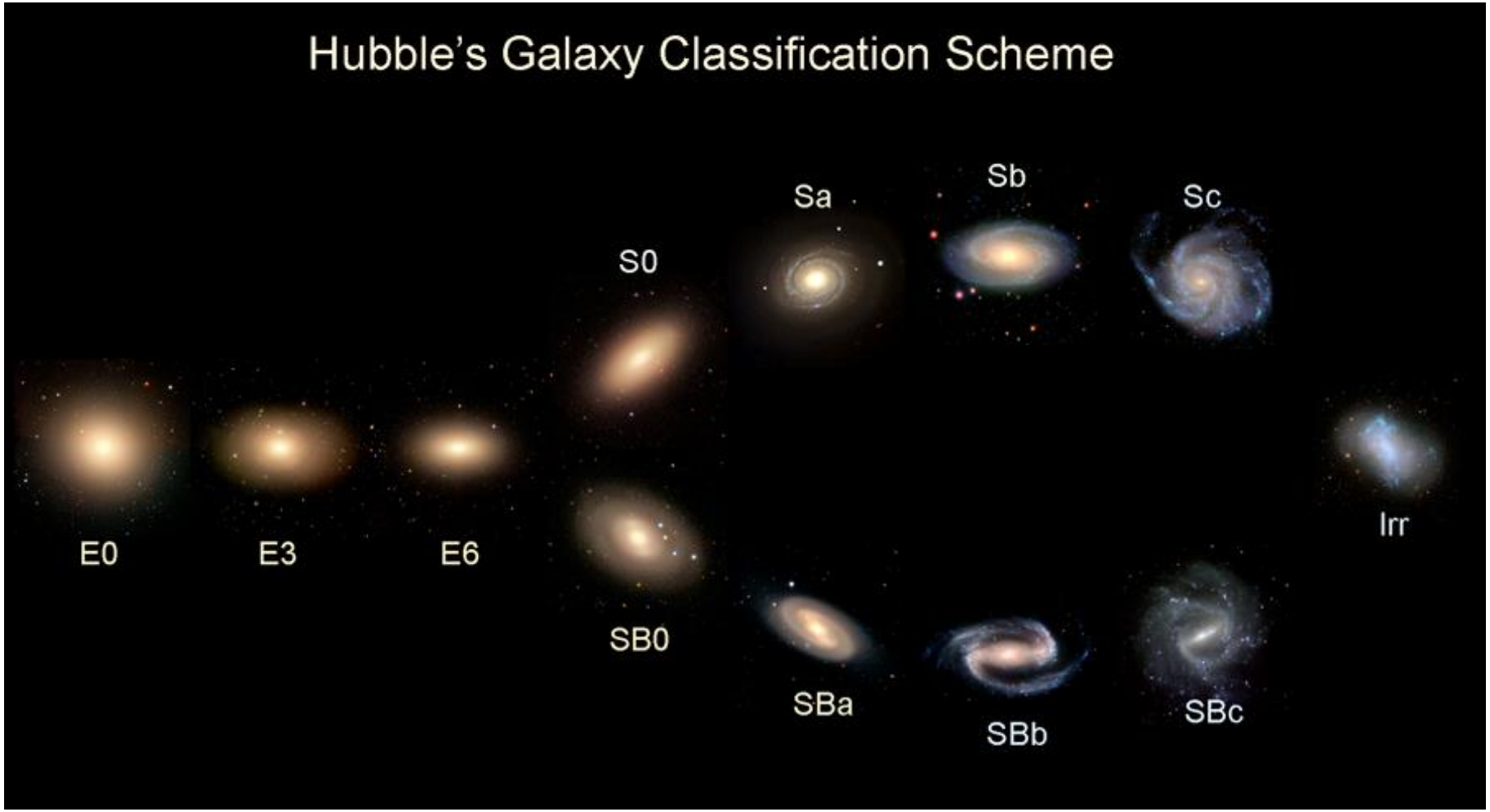
- The physical characteristics of galaxies
- Galaxy scaling relations
  - Tully-Fisher
  - Faber-Jackson and the fundamental plane
  - Black hole mass



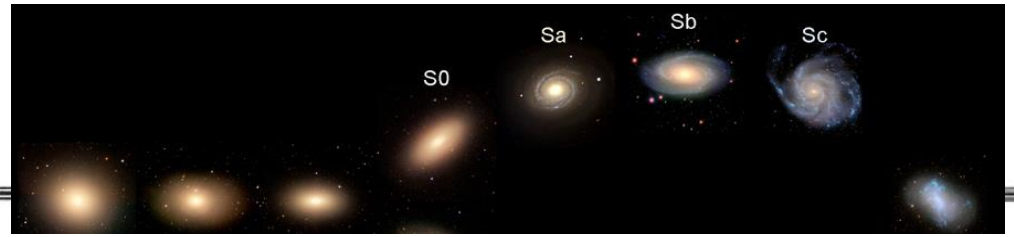
The zoo of galaxies, arranged in colour and luminosity. Note that the brightest galaxies tend to be rather yellow.

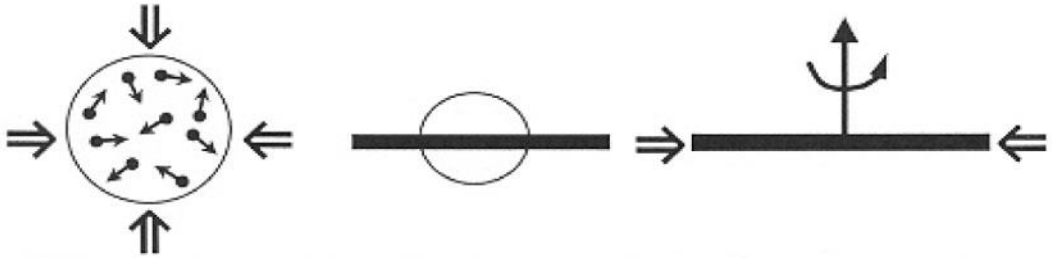
Galaxy types

The classic classification of galaxies is the *Hubble tuning fork* – based on morphology.
Does this make any sense in terms of the physical properties of galaxies?



Galaxy types





Ellipticals	Lenticulars	Spirals	Irregulars
spheroid		**disk**	
"thermal" support		centrifugal support	
slow rotation		rapid rotation	
redder color		**bluer color**	
old pop's		old+young pop's	
higher Z's		lower Z's	
coeval formation		extended formation	
less gas		more gas	
low SFR		high SF	
compact		extended	
more massive		less massive	
20%	10%	70%	

redder

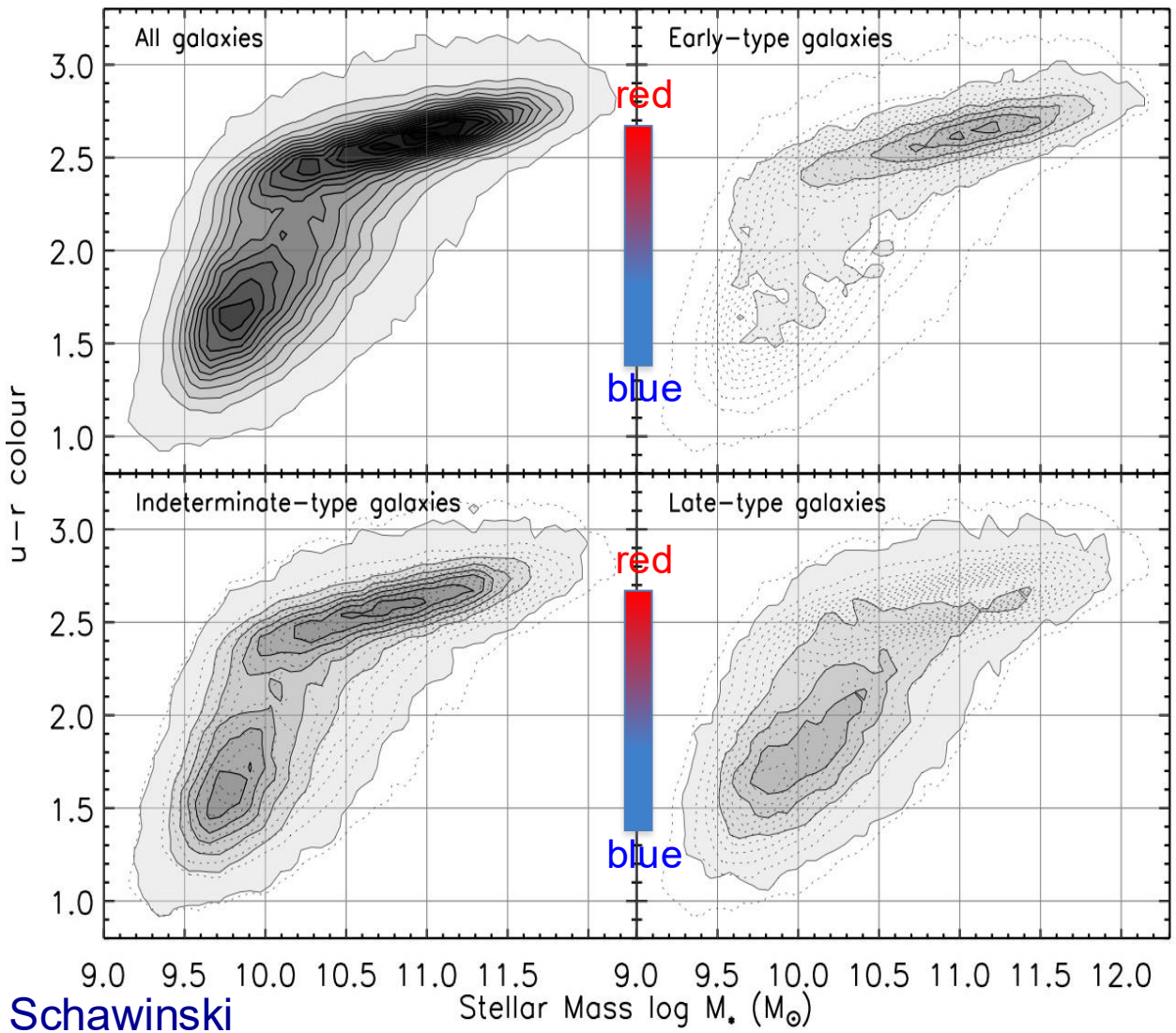
bluer

Although the distinction between barred and unbarred spirals does not seem to be fundamental. Many galaxy properties do progress in a rather well-defined way along the sequence from late to early-type galaxies.

We would like to understand these patterns in galaxy properties in terms of the way they have formed.

Avila-Reese (2006)

Galaxy types

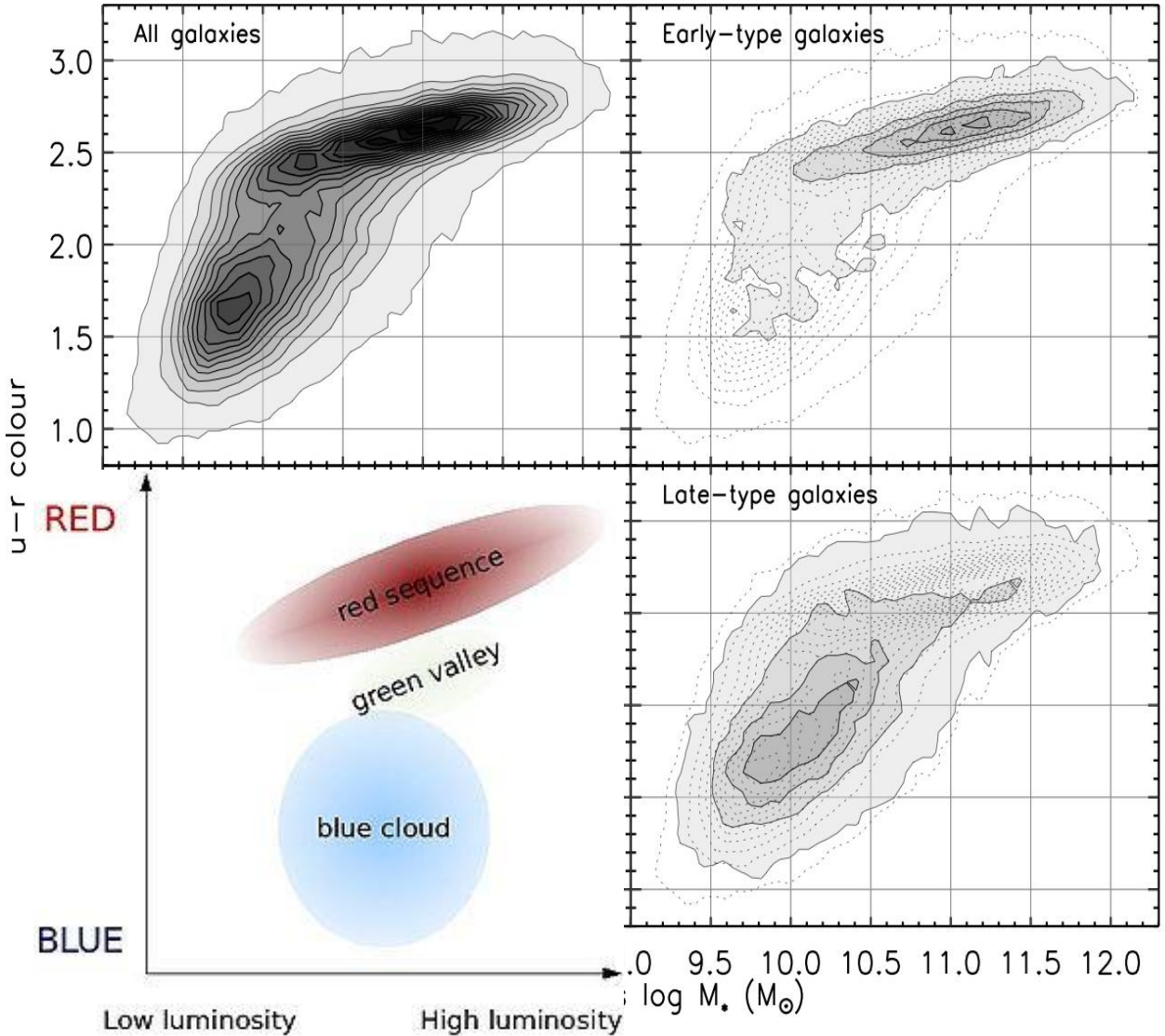


Schawinski

Large galaxy surveys such as SDSS (the Sloan Digital Sky Survey) and 2dF have made available massive datasets to study galaxy properties. Since galaxy morphology is hard to quantify, it is more convenient to use colour.

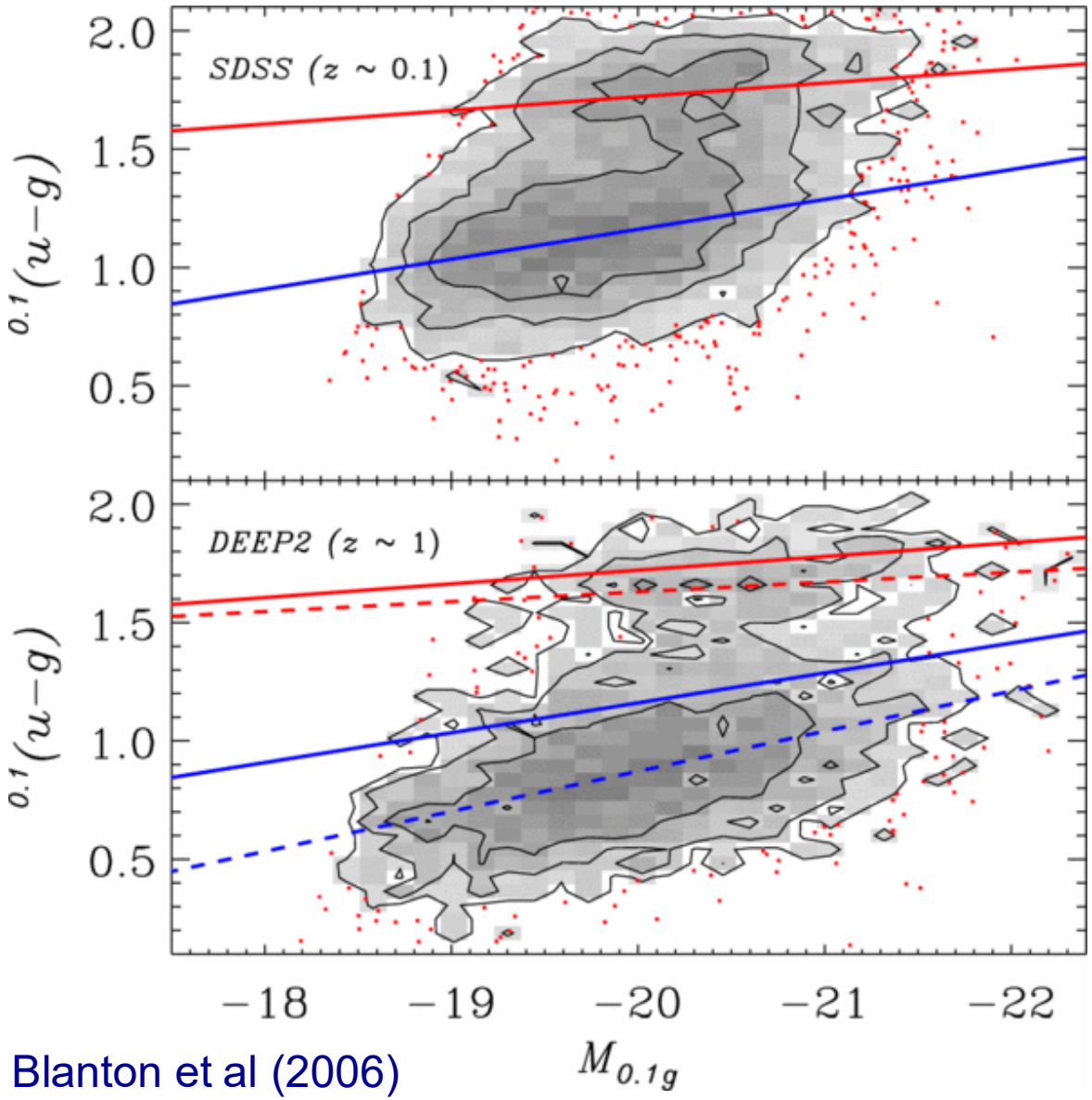
In practice, these is a fairly close correspondence between the two.

Galaxy types



The two main concentrations of galaxies seen in the colour-luminosity plane are known as the red sequence and blue cloud. Between these is a region now called the green valley, which is more sparsely populated with galaxies.

Galaxy types



Blanton et al (2006)

The same pattern of red sequence+blue cloud is seen in galaxies at *z*~1. The main difference being that both (but especially the blue cloud) have somewhat bluer intrinsic colours at high redshift.

We will see below that *environment* has a strong influence on galaxy properties.